ÖREBRO UNIVERSITY SCHOOL OF BUSINESS · ECONOMICS

# The individual report: instructions

Economics Research and Communication (NA310G), the applied part

Autumn 2026 · Magnus Lodefalk

In the computer labs, you learn how to work with panel data to perform microeconometric analyses. In the individual report, you then apply the knowledge, skills and methods from the lectures, the exam preparation and the labs. You show that you can comprehend at least one research article in economics, take a research question, make a quantitative analysis of it, communicate a correct interpretation of your results, and discuss critically both the economic and the methodological issues at hand.

Everything about the labs, the data, AI and the help desks is also on the course site, **magnuslodefalk.com/na310g**. Where this document and the site differ, this document applies.

**At a glance**

- **Deadline:** Sunday 8 November 2026, 16:00, one PDF by e-mail (subject `NA310G: Assignment`), named with your code (`2HA9JR-FINAL.pdf`).
- **Length:** at most 2,500 words; the appendix holds your do-files, final log, verification line and AI declaration.
- **Your question and data:** personal, downloaded with your code from the course site.
- **Labs:** attend at least two of three (22, 26, 28 October); help desks 23 October to 6 November.
- **AI:** to learn, never to do the analysis or write the report; a three-line declaration.

## 1 The computer labs

There are three labs (22, 26 and 28 October). **Attend at least two, preferably Lab 1 and Lab 2**; participation is required before you can submit the report. In the labs you will:

- learn to work with panel data in Stata, in a project folder with numbered do-files that anyone can rerun;
- estimate pooled OLS, fixed and random effects and instrumental variables, and run the related tests;
- produce the tables a thesis or an academic article needs, from your own do-files.

Outside the labs you also learn to manage data in Stata (merging and collapsing), with short videos on the course site. Help desks run from 23 October to 6 November (times on the course site). Your project folder (the starter) and your personal data come before Lab 1, via the course site.

## 2 The report in one paragraph

You write an individual research report of **at most 2,500 words** (tables, references and the appendix not counted). You summarise one article, present and discuss descriptive statistics on your own dataset, and present and discuss your own panel regressions that answer your research question. The research question comes from the literature on how technology, and AI in particular, affects

firms, occupations and jobs. You get it in a broad sense: your outcome ("y"), the key explanatory variable ("x"), the required controls, your industry and your geography. See *Your research question* below. **Deadline: Sunday 8 November 2026, 16:00.**

## 3 Literature and other sources

The starred articles are mandatory to read carefully, and you summarise **one of them** in your report. The #-items are mandatory to browse; the rest are optional. All are linked from the course site.

- *Engberg, E., Görg, H., Lodefalk, M., Javed, F., Längkvist, M., Monteiro, N. P., Kyvik Nordås, H., Pulito, G., Schroeder, S. and Tang, A. (2024). AI Unboxed and Jobs: A Novel Measure and Firm-Level Evidence from Three Countries. IZA Discussion Paper No. 16717. (The source of the AI exposure measure, DAIOE, in your data.)
- *Engberg, E., Hellsten, M., Javed, F., Lodefalk, M., Sabolová, R., Schroeder, S. and Tang, A. (2025). Artificial Intelligence, Hiring and Employment: Job Postings Evidence from Sweden. *Applied Economics Letters*, early online.
- #Hellsten, M. (2025). Return to AI Skills. Manuscript.
- #SCB (the most recent version). *Näringslivets användning av AI*. Statistics Sweden, report.
- Felten, E., Raj, M. and Seamans, R. (2021). Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses. *Strategic Management Journal*, 42(12).
- Engberg, E., Koch, M., Lodefalk, M. and Schroeder, S. (2025). Artificial Intelligence, Tasks, Skills and Wages: Worker-Level Evidence from Germany. *Research Policy*, early online.
- Babina, T., Fedyk, A., He, A. and Hodson, J. (2023). Firm Investments in Artificial Intelligence Technologies and Changes in Workforce Composition. SSRN, manuscript.
- Acemoglu, D., Autor, D., Hazell, J. and Restrepo, P. (2022). Artificial Intelligence and Jobs: Evidence from Online Vacancies. *Journal of Labor Economics*, 40(S1), S293–S340.
- Lodefalk, M. (2024). *Artificiell intelligens och jobben*. Ratio.

Other sources: SCB statistics on firms' AI use (Statistikdatabasen > Näringsverksamhet > IT-användning i företag > AI) and on the population's use of generative AI (Statistikdatabasen > Levnadsförhållanden > Befolkningens it-användning); Eurostat statistics on AI use in enterprises; the AI-Econ Lab's papers and its AI Monitor (ai-econlab.com).

## 4 The data

You work with a firm-year panel of the large Swedish firms (at least 250 employees in some year), **2018–2024**. It combines three kinds of data that researchers use to study AI and work:

- **firm accounts** from **Serrano**, the Swedish House of Finance's database of company financials;
- **job advertisements** from Platsbanken (Arbetsförmedlingen, via JobTech), counted per firm and year, with the ads that ask for AI skills identified by the AI-Econ Lab's AI Monitor;
- **occupational AI exposure** (DAIOE; Engberg et al., 2024) and O*NET ability scores for the occupations the firm advertises for (SSYK 2012), in a separate occupation file that you merge yourself.

The file you get is a **synthetic twin** of this research panel: the variables you need, with the real names, units and structure, and statistical properties close to the real data, but every firm is simulated. The quirks of real accounts are kept on purpose (costs stored as negative numbers, firms with zero employees, outliers, missing values), because handling them is part of the job. **Your dataset is personal**: you download it with your personal code (page *Your data* on the course site), so your numbers differ from your classmates'. The codebook in the download describes every variable.

Never upload the data, or any file made from it, to an AI tool (see *AI in the labs and the report* below). For your bachelor thesis you can use the real Serrano data, free for ORU students (course site, *Data*).

## 5 Your research question

There are many questions about whether and how AI affects firms. Does a firm that adopts AI perform better, in sales or in productivity? Does it pay its workers more, because it grows or upskills, or less, because it automates? Does it restructure, hiring for new tasks while shedding others? This term, each of you explores one such question. Your personal download contains a file `research_question.txt` with your outcome, industry and geography.

**Key explanatory variable (the treatment):** `ai_firm`, which is 1 if the firm posted at least one job ad asking for AI skills that year and 0 otherwise (also 0 for firm-years without any ads). Whether the firm posted 2 or 20 ads, and whether one or all of them asked for AI, `ai_firm` is 1. Most firm-years are 0. Note that 0 mixes two kinds of firm-years: those that advertised but not for AI, and those that did not advertise at all (`has_ads` tells them apart). That is worth a sentence in your discussion.

**Outcome:** one of four, given to you:

1. Salaries paid (natural logarithm).
2. Value added per employee (natural logarithm): labour productivity, how efficiently the firm uses labour to create value.
3. Net sales (natural logarithm): the firm's ability to generate revenue.
4. Job vacancies per employee: generate it by dividing the number of vacancies by the number of employees. A way to capture whether the firm expands or restructures.

**Required controls:** firm size (number of employees, natural logarithm, for outcomes 1–3; net sales, natural logarithm, for outcome 4), the capital stock (book value of buildings and land plus machinery and equipment, summed before taking the natural logarithm), and year fixed effects. You may add one or at most two more, motivated by theory or a paper you read.

**Industry:** one of four, given to you (SNI 2007; learn its hierarchy at scb.se):

- Manufacturing: section C;
- Professional services: sections J + K + M;
- Other services: sections F + G + H + I + L + N + S;
- All industries.

**Geography:** one of three, given to you (counties by SCB's county code):

- all of Sweden;
- all of Sweden except Kalmar and Västerbotten counties;
- all of Sweden except Blekinge and Norrbotten counties.

About variables in general: be clear about how you treat negative values. If all values are negative, it is a cost in the firm's books, and you flip the sign. If only some are negative or zero (a loss, a firm with no employees, no capital), you cannot take the logarithm of them: decide what to do, report how many firm-years it concerns, and say why. There is no single right answer; an unexplained one is wrong. Tell Stata whether a right-hand-side variable is continuous or categorical with factor notation (`i.`), and use it for lags too (`L.`).

## 6 AI in the labs and the report

These rules cover the computer labs and this report, the part of the module taught by Magnus Lodefalk. For Daniela Andrén's lectures and the written exam, follow the course guide and her instructions.

You are allowed, and encouraged, to use generative AI tools in the learning process and to improve the language of your report. **No student is allowed to use an AI tool to execute the analysis or write the report**: that is plagiarism, and disciplinary action follows. Where the line goes: asking what an error means, how a test works, or for suggestions on the language of a paragraph *you wrote* is fine; letting a tool write paragraphs, interpret your results or produce more than a line or two of your analysis code is not. Use that is within the rules and declared has no consequence for your grade. If you are unsure, ask (at Lab 1, a help desk, or by e-mail) before you act. The rules, a tutor prompt and examples are on the course site, page *AI rules*. In short: try first yourself; use AI to explain, debug or suggest; test what it gives you; explain every line; mark lines that came from AI with `* AI:` in your do-files. Never upload the course data.

**The AI declaration** (in the appendix, three lines, modelled on what economics journals require):

1. **Tools:** which tools you used (name and version), or "none". The spelling and grammar check built into Word need not be listed; any generative AI tool, also when used for language, is listed.
2. **What for and how:** for example, explaining an error message, checking your reading of the Hausman test, improving the language of section 2; and which lines in your do-files carry `* AI:`.
3. **Responsibility:** "I reviewed everything these tools produced, and I take full responsibility for the content of this report, including every line of code and every number."

No prompt logs.

## 7 Formalities and submission

No form of cheating, including plagiarism, is allowed; this includes attempts and helping someone else to cheat. It is your responsibility to follow the rules of Örebro University; if you are uncertain what counts as cheating, contact juridik@oru.se.

- **Your code, never your name.** Use your personal code on the title page ("Code: 2HA9JR"), and in the file name (`2HA9JR-FINAL.pdf`). No names or personal identity numbers in the report, do-files or logs.
- **Submit one PDF by e-mail by Sunday 8 November, 16:00**, to magnus.lodefalk@oru.se with the subject `NA310G: Assignment`. The PDF is the report followed by the appendix ("bilaga"): save the report as PDF from Word (`2HA9JR.pdf`), print each do-file and log to PDF from Stata's editor or viewer, and combine the files (Preview on a Mac, Acrobat or a free PDF tool on Windows; video V9 shows it). Name each do-file's PDF after the do-file plus `-do` and each log's PDF plus `-log` (`0_settings-do.pdf`, `0_settings-log.pdf`), so you can tell them apart when you combine them. Name the combined file with your code and `-FINAL`, for example `2HA9JR-FINAL.pdf`, and send it from your student e-mail. Your sent e-mail is your receipt: keep it. If you send a corrected file before the deadline, the last one counts.
- **The report has** a title page (template below), page numbers on every page, at most 2,500 words, and Harvard references. The cap counts the text of sections 1 to 8, including the summary of the article; it does not count the title page, tables with their notes, references or the appendix.
- **The appendix has** any additional tables and figures; your do-files; the final log (not the one with every trial and error) with your brief comments; the **verification line** ("I deleted

`data/processed` and `output`, ran `master.do`, and got N = … and b = … as in Table …"); and the **AI declaration**. The first lines of a log show your folder path: if it contains your name, delete that line in the PDF.

- **See an example:** a complete example report, on a question none of you has (the lab demo question, chosen to show the methods rather than to find anything about AI: do AI-adopting firms have higher financial expenses?), with notes on why each part is there, is on the course site, page *The report*, from Monday 19 October. Use it to see what a complete report looks like; your question, sample and numbers are your own. **Text taken from the example report is treated like any other copied text.**
- **Check before you hand in:** `report_checklist.pdf` in your project folder lists the things that most often sent a report back for a revision in earlier terms.

**After submission.** Within about two weeks you get your result. **Some students may be invited to a short follow-up conversation (about 10 minutes) about their own code and numbers**: for example, reports near a grade boundary, reports where something needs explaining, or a few chosen at random, so being invited says nothing about you. It takes place before your grade is final and is not graded on its own; it confirms that the report is yours. You will be asked what a few lines of your code do and what a number in your table means, with your report in front of you. Nobody expects you to recite Stata from memory; if you wrote the code yourself, it is easy.

If you have submitted on time and your report is graded as a conditional pass (it is close but something essential is missing or wrong), you have one possibility of a major revision and one additional minor revision, each with written feedback and a stated deadline. If you still do not pass after these two submissions, you will have to come back and do an assignment the next time the course is given.

**The class figure.** After grading, the estimated AI coefficients of the whole class may be shown, anonymously, in one figure to next term's students: how much the answer to one question varies with reasonable choices of outcome, industry and geography. It never affects your grade, no code, name or text is shown, and estimates are grouped so that no single student's number can be traced.

## 8 What earns a pass, and what earns a VG

**Pass (G):** every part of the template is there; the data and sample are described; pooled OLS and fixed or random effects are estimated and presented correctly; the main coefficient is interpreted correctly (sign, size, unit, significance, ceteris paribus); the Hausman test (or its robust version) and the two tests in section 7 are run and read correctly; the conclusion answers the question; the formalities and the appendix are complete.

> **Misreading the main coefficient or a test is the most common reason a report is not passed.** Write the null hypothesis before the verdict, and the coefficient's sign, size, unit and significance in words.

**Pass with distinction (VG):** in addition, a well-founded hypothesis from the article and theory; a careful discussion of what the estimates can and cannot show (selection, reverse causality, what fixed effects remove and what they do not); an instrument argued for both conditions, and estimated if it is credible, or a reasoned case that no credible instrument exists; and policy conclusions that follow from the results. An insignificant result or a weak instrument does not stand in the way of a VG: the quality of the reasoning does.

## 9 Template for the report

Show that you comprehend at least one research article in economics, that you can make a quantitative analysis of an economic question in that literature, interpret the results, discuss critically both the economic and the methodological issues, and communicate the results in writing.

**Title page.** Örebro University · Örebro University School of Business · Economics, Economics Research and Communication (NA310G) · Autumn 2026 · TITLE · Code: your code.

1. **Introduction and research question.** Your research question, why it is interesting and how it relates to previous literature, the purpose of the study, and a short outline including the main result.
2. **Summary of previous studies.** Summarise the article you read: its question, why it matters, its contribution, theory, data, empirical strategy, results and conclusions, for a fellow economics student; about one page (at most two). Refer as in "Engberg et al. (2025) study this, using job ads. They find that …", or "(Engberg et al., 2025)"; titles go only in the reference list.
3. **Theory and hypotheses.** A theoretical framework supported by earlier studies (with references), and one or more testable hypotheses that correspond to the research question, as a null and an alternative.
4. **Data.** Source, type (panel), years and population; whether the data suit the empirical strategy; the sample selection step by step, from the number of observations in the original file to your final sample. Define all variables in an appendix table. Present descriptive statistics of all variables in your final analysis (N, mean, **median**, standard deviation, minimum and maximum) and their correlations, and interpret them. Use English decimal points (0.25) and thousands separators (1,250).
5. **The econometric approach.** A full specification of your regression equation. Use pooled OLS and fixed or random effects. Consider whether IV estimation would be motivated and why or why not, and which conditions an instrument must fulfil. For a VG, propose an instrument and argue both conditions (and run it if it is credible), or give a reasoned case that no credible instrument exists; see *What earns a pass, and what earns a VG*.
6. **Results.** All specifications in one properly formatted table (at least two columns), with a clear caption and a note (standard errors, stars, sample). Build tables from your do-files (`dtable` and `collect`, as in the labs, or `etable`), never by hand. Under the table, interpret every coefficient: magnitude (with units, and per cent for a dummy in a log outcome), statistical significance and the level used, ceteris paribus. Report the R-squared that fits each model (within for fixed effects, overall for random effects). Test whether the random-effects assumption holds (Hausman, or its robust version from Lab 2) and conclude. (Hausman needs the models without clustered errors, for the test only; your table keeps clustered errors.) Test whether your added controls are jointly significant (`test` or `testparm`, Lab 2). Discuss whether controlling for firm heterogeneity matters.
7. **Tests for your preferred specification.** Heteroskedasticity and multicollinearity, each with its number and what you did about it.
8. **Concluding remarks.** Summarise the findings in relation to the purpose and the question; policy implications; whether the results generalise; the key identification problems; other problems for future work.
9. **References** (Harvard).

**General advice.** Association is not the same as a causal effect; the gold standard is a randomised controlled trial. Without one, avoid causal language unless your design earns it: write "associated with", "linked to", "correlated with". For industry heterogeneity, include industry fixed effects with factor notation (`i.industry`), not the industry code as a number; with firm fixed effects they drop out, because a firm's industry does not change. Present tables with three horizontal lines at most, the caption above (*Table 1. Descriptive statistics*), notes directly below.